OOAD and Agile

Learning Outcome –Introduction to CRC cards,

- Describe CRC cards, and the steps used to draw them

Entity Class Using CRC Analysis

Class Name :BankAccount	
Responsiblities Maintain Balance Candidate Class	Collaborators Bank Other classes that may Collaborate with BankAccount Class to realize the responsibility

Entity Class
Using CRC Analysis

CRC Card	
Class:	StudentRecord
Responsibilities:	Collaborators:
contains description of student	Description
has a collection of Assignments	Assignment
changes grades and Curve Status for assignments when asked by Gradebook	GradeBook, Assignment

Responsibilities and Collaborations

- Responsibilities
  - Knowing
  - Doing
- Collaboration
  - Objects working together to service a request
- Object oriented systems are made up of objects working together to provide functionality
- CRC cards can help to form a “bridge” from structural model to behavioral model

The essentials of CRC cards

- Extremely simple technique
- Rather than using diagrams to develop models use 4 x 6 index cards
- Rather than indicating attributes and operations on the cards, they identify *responsibilities* and *collaborators*

CRC Cards

- On the index card, record the following information:-
  - **the name** of the class, at the top
  - **the responsibilities** of the class, on the left-hand side
  - **the collaborators** of the class, which help to carry out each responsibility, on the right-hand side of the card
- Many templates exist

Example Simple CRC card

Student	
Responsibility	Collaborators
Complete Assessment	Assessment
Submit Assessment	School office
Attend Lecture	Lecture
Attend Seminar	Seminar

Exercise 1

- Think about the working of a PUB and the interactions between
  - Bar person
  - Customer
- Draw CRC cards for these two classes
- This may involve you identifying other classes
- This should take about 10 minutes

Exercise 1 – One Possible Answer

Bar Person	
Responsibility	Collaborators
Take orders from customers	Order Customer
Serve orders	Order Product
Take payments and give change	Till
Check stock	Stock

Exercise 1 – One Possible Answer

Customer	
Responsibility	Collaborators
Give orders	Bar Person Order
Receive orders	Bar Person Order Product
Make payments	Bar Person

What have we seen

- Identification of “new” classes
  - Order
  - Product
  - Till
  - Stock
- How classes interact to provide functionality
- A feel for behavioural aspects of the system
- Each of these would also need a CRC card

CRC Good Practice

- If a class has lots of responsibilities :-
  - Can describe them more concisely, and, if not
  - Would be better to split the responsibilities and have two or more classes
- Too many responsibilities:
  - Bad- low cohesion
- Too many collaborators:
  - Bad -high coupling
- Using CRC cards should help to identify and fix both these faults in a model

Why use CRC cards

- At the start, you may not know which other classes a class has to collaborate with in order to fulfill a responsibility
- Using CRC cards facilitate this
- If an object of one class collaborates with an object of another class, it does so by using operations of the other
  - A class has a relationship with each of it’s collaborators
- Using CRC cards helps to establish in which direction associations should be navigable

CRC cards in practice (1 of 2)

- One of the main benefits of CRC cards is that they encourage animated discussion among developers
- Especially effective when walking through a Use Case to see how the classes are going to implement it
- Developers pick cards as each class collaborates in the Use Case

CRC cards in practice (2 of 2)

- Stops thinking about classes as dumb data holders and moves thinking towards understanding the higher-level behavior of each class
- CRC cards work on '*object*' communication (not classes)
- Can provide a draft class model
- Apart from the technical benefits, stimulates
  - Team discussion and hence team spirit

Exercise 2

- Consider a Library.
- A library has Borrowers, Books Copies of Books and Reservations, LendingDetails,
- Borrowers can borrow copies of books and make reservations, LendBooks
- Draw CRC cards for the library

Exercise 2 – One possible answer

Borrower	
Responsibility	Collaborators
Borrows books	Book, Copy,Member, LendingDetails
Return books	Book, Copy, Member, BorrowedCopy
Make reservations	Book,Member ReservationDetails

Exercise 2 – One possible answer

Book	
Responsibility	Collaborators
To have copies	Copy
To be borrowed	Borrower Copy
To be reserved	Reservation Borrower Copy

Exercise 2 – One possible answer

Copy	
Responsibility	Collaborators
Provides book info	Book
Provides borrower info	Borrower
Provides reservation info	Reservation

Exercise 2 – One possible answer

Reservation	
Responsibility	Collaborators
To create a reservation for a book	Book Borrower, ResearvationDetails
To hold copy	Copy Book

Stereotypes and Classes cont..

Control Class:

- Sequencing behaviour specific to one or more use cases.
- There is typically one *control* class per use case.
- Co-ordinates the events needed to realise the behaviour specified in the use case.

Eg. Running or *executing* the use case.

Finding Controller Classes

- Simple behavior can often be distributed between Boundary or Entity classes
- Consider more complex behavior of the system as described by the use cases.
- Work out how these behavior should be partitioned among the analysis classes.
- Control classes process messages from an interface class and respond to them by sending and receiving messages from the entity classes.